Klarrow

Evidence-reconciliation workbench

AI ACADEMY SOUTH AFRICA · PITCHATHON

Evidence reconciliation for food-traceability investigations

Klarrow helps an operator assemble and challenge fragmented supplier, production and dispatch records - with source grounding, human decisions and explicit uncertainty.



Human review. Source grounding. Explicit uncertainty.

Prototype demonstration · 18 October 2026 · Run 4 controlled synthetic scenario

THE INVESTIGATION MOMENT

A supplier alert starts a document hunt, not a clean database query.

01

Which batches used the lot?

Production sheets, batch records and stock ledgers rarely share one identifier.

02

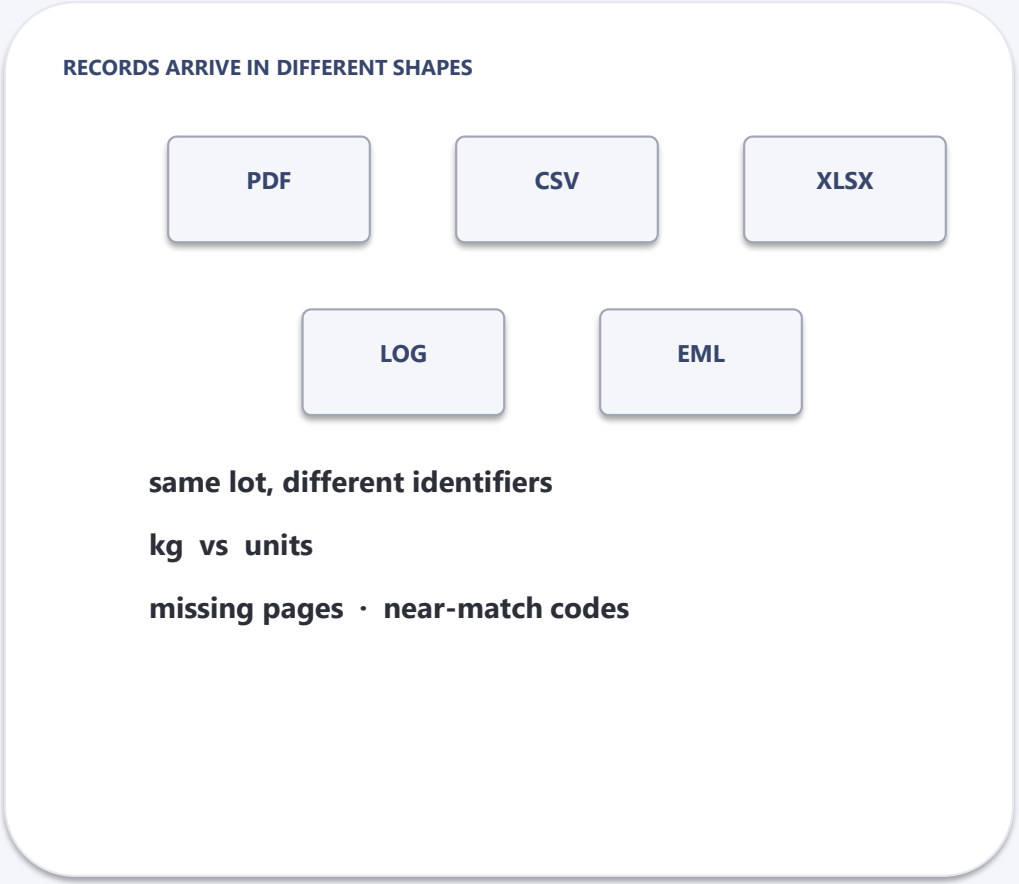
Which customers received them?

Dispatch notes and invoices sit across warehouses, carriers and spreadsheets.

03

Where does the evidence stop lining up?

Quantities disagree; records are missing; typos look like real lot codes.



The operator must assemble, challenge and document the evidence trail. Klarrow is designed for that moment - not to decide food safety.

THE WORKBENCH

One workbench: ingest, ground, review, trace, export.

1 Ingest › 2 Extract › 3 Ground › 4 Review › 5 Trace & export



SUPPORTED INPUTS

Text · CSV/TSV · Excel · EML · logs · text-selectable PDF

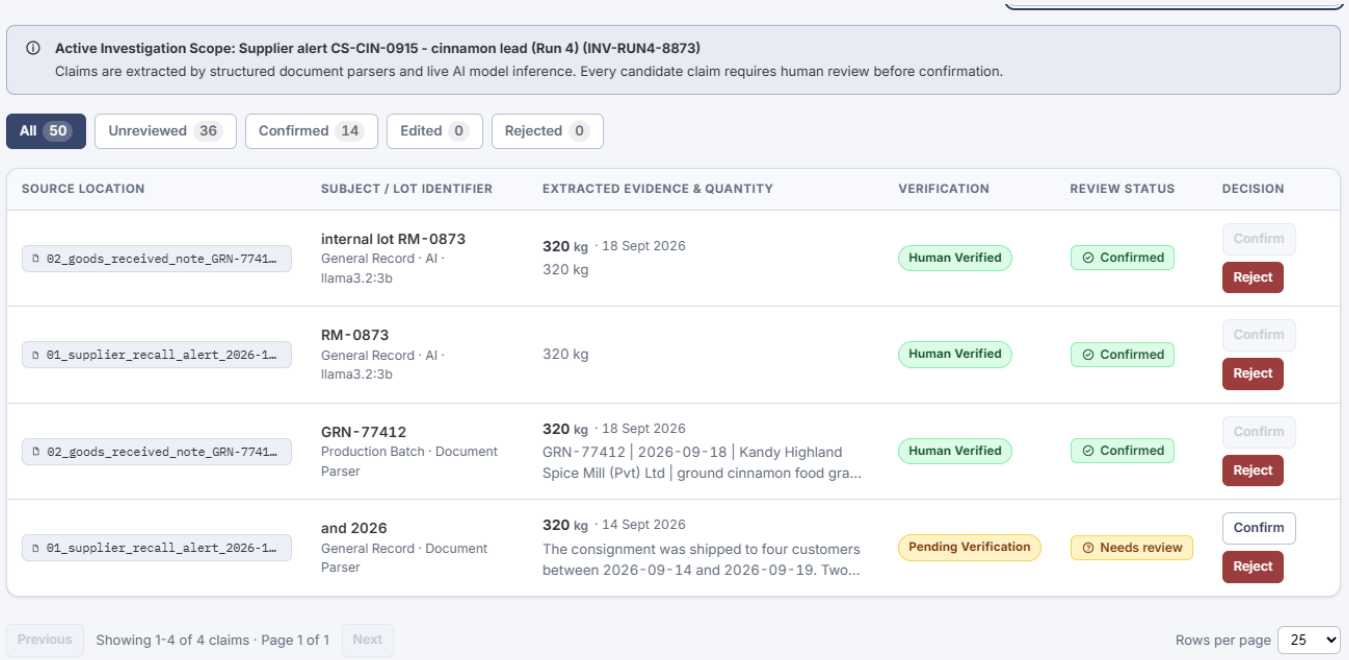
HONEST BOUNDARY

Images and scanned PDFs are retained but not OCR'd. The prototype does not read handwriting or photographs.

Klarrow sits beside the system of record. It does not replace the ERP.

THE AI ROLE

Meta Llama proposes candidate facts. Deterministic code and people decide.



LOCAL META LLAMA 3.2 3B

Candidate fact extraction from supported text. No cloud model call in this run.

DETERMINISTIC PARSERS

Structured rows, lots, quantities and trace-link proposals. Graph traversal is code, not the model.

HUMAN REVIEWER

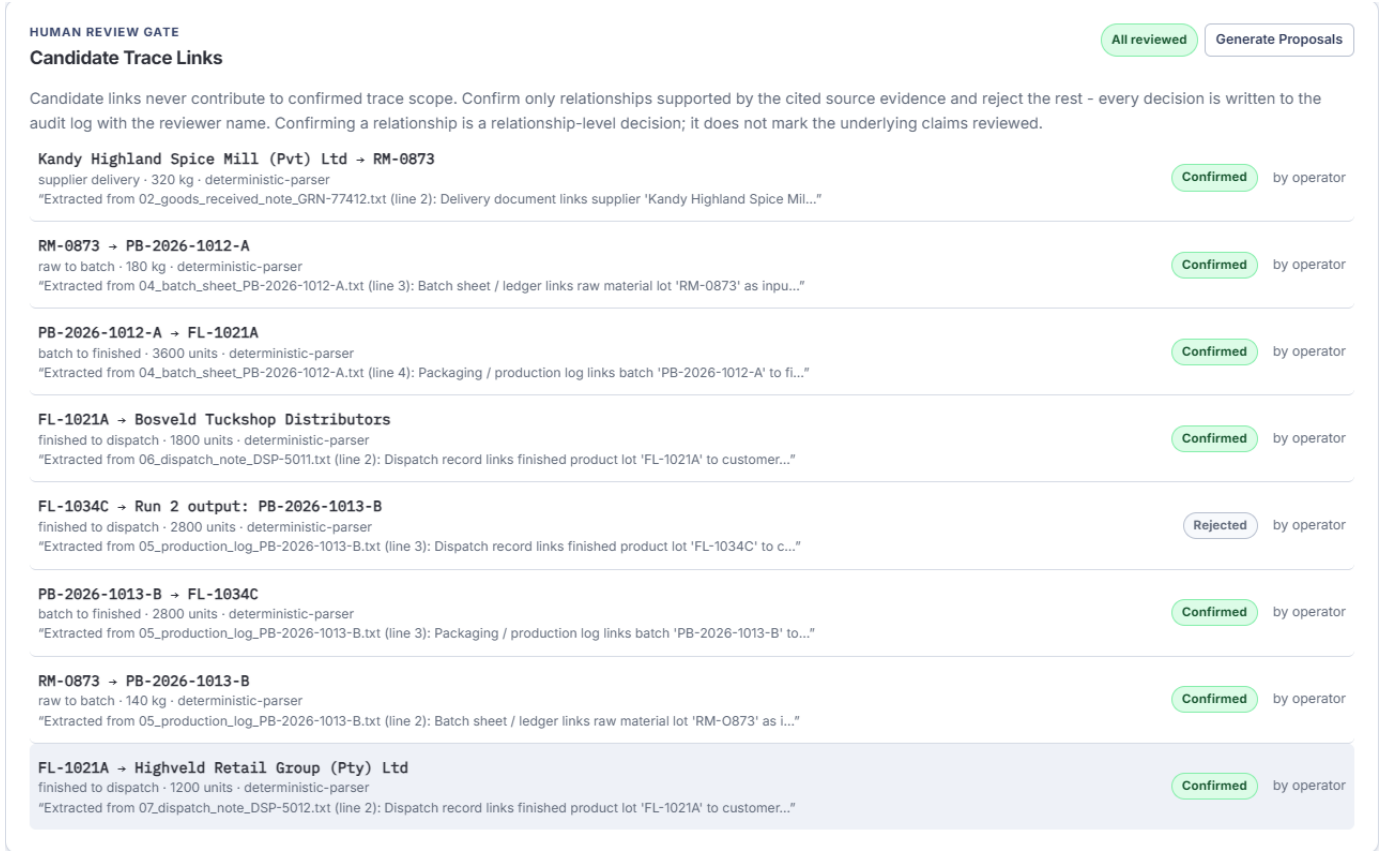
Confirms, rejects or edits with a reason. Every decision is written to the audit trail.

15	35	4	3	8
AI rows	parser rows	value-in-quote	adjacent context	identifier-only

Run 4 · one authored seven-document scenario · 15 AI rows contain duplicate values; this is not 15 unique discoveries. No unique AI-only raw token was established.

THE HUMAN GATE

A relationship is not confirmed just because a model proposed it.



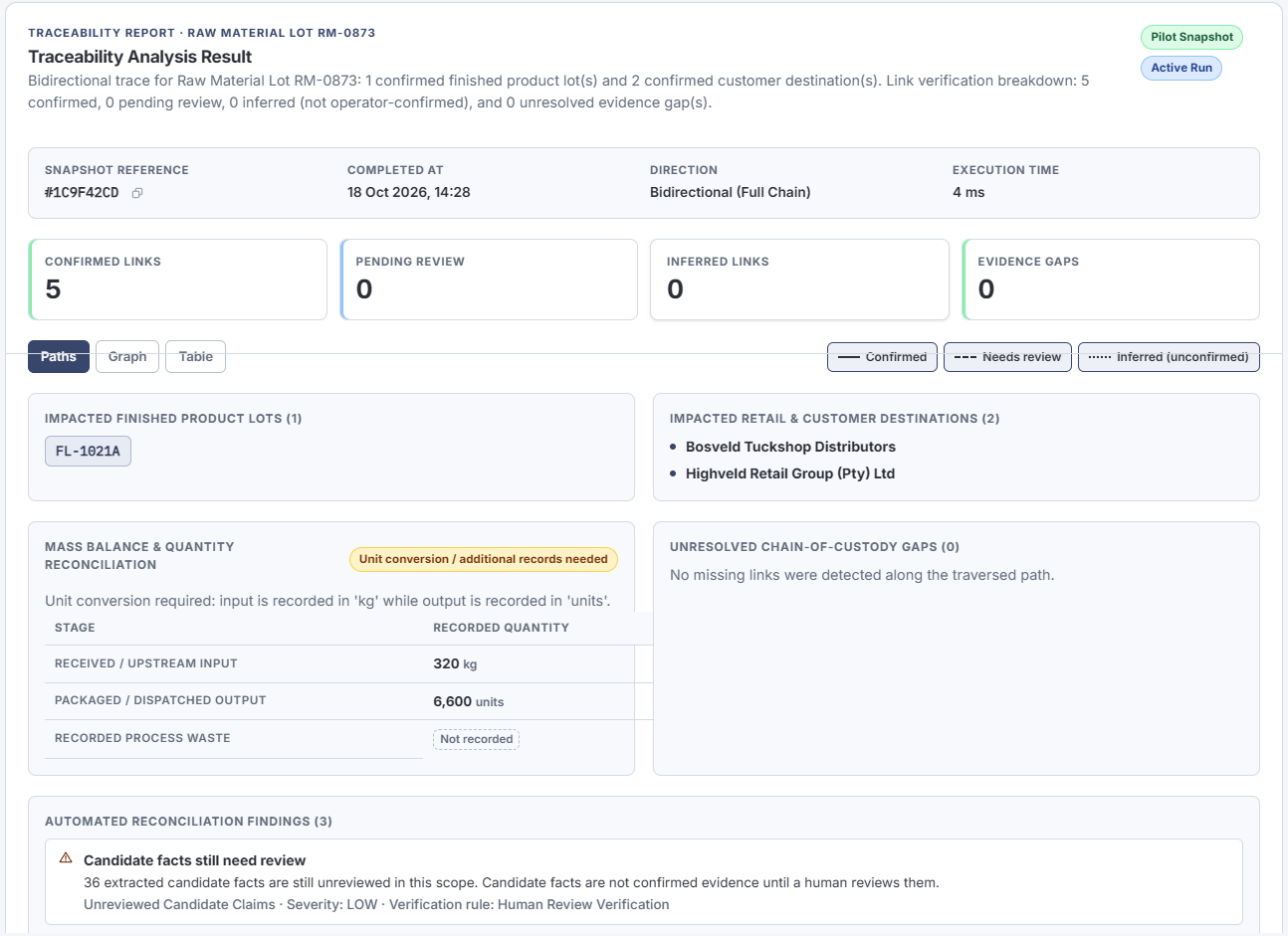
RUN 4 REVIEW OUTCOME

- 8 relationship proposals.
- 7 confirmed by the operator; 1 rejected.
- The rejected link is a false-positive dispatch phrase: "Run 2 output: PB-2026-1013-B".
- RM-O873 / RM-0873 alias deferred, not silently merged.
- Supporting facts reviewed for confirmed links; 36 facts remain unreviewed.

Confirmation records a human decision. It is not independent proof of truth.

The false positive is a feature of the demo, not a hidden failure: the gate caught it before it entered confirmed scope.

The trace separates confirmed scope from possible exposure.



TRACEABILITY REPORT · RAW MATERIAL LOT RM-0873

Traceability Analysis Result

Bidirectional trace for Raw Material Lot RM-0873: 1 confirmed finished product lot(s) and 2 confirmed customer destination(s). Link verification breakdown: 5 confirmed, 0 pending review, 0 inferred (not operator-confirmed), and 0 unresolved evidence gap(s).

Pilot Snapshot

Active Run

SNAPSHOT REFERENCE	COMPLETED AT	DIRECTION	EXECUTION TIME
#1C9F42CD	18 Oct 2026, 14:28	Bidirectional (Full Chain)	4 ms

CONFIRMED LINKS	PENDING REVIEW	INFERRED LINKS	EVIDENCE GAPS
5	0	0	0

Paths | Graph | Table

— Confirmed --- Needs review ····· Inferred (unconfirmed)

IMPACTED FINISHED PRODUCT LOTS (1)

FL-1021A

IMPACTED RETAIL & CUSTOMER DESTINATIONS (2)

- Bosveld Tuckshop Distributors
- Highveld Retail Group (Pty) Ltd

MASS BALANCE & QUANTITY RECONCILIATION

Unit conversion / additional records needed

Unit conversion required: input is recorded in 'kg' while output is recorded in 'units'.

STAGE	RECORDED QUANTITY
RECEIVED / UPSTREAM INPUT	320 kg
PACKAGED / DISPATCHED OUTPUT	6,600 units
RECORDED PROCESS WASTE	Not recorded

UNRESOLVED CHAIN-OF-CUSTODY GAPS (0)

No missing links were detected along the traversed path.

AUTOMATED RECONCILIATION FINDINGS (3)

⚠ **Candidate facts still need review**

36 extracted candidate facts are still unreviewed in this scope. Candidate facts are not confirmed evidence until a human reviews them.

Unreviewed Candidate Claims · Severity: LOW · Verification rule: Human Review Verification

RUN 4 SNAPSHOT

Confirmed

FL-1021A → Bosveld + Highveld

Outside this trace

FL-1034C stays outside the RM-0873 trace while the RM-O873 alias is deferred.

Possible scope

Empty in this snapshot.

Mass balance

Insufficient: 320 kg input vs 6,600 units output. No conversion invented.

3 open findings

Not a recall decision.

THE EVIDENCE BOUNDARY

What this run supports - and what it does not.

OBSERVED

- End-to-end workflow on one authored scenario.
- Local Llama candidate facts; deterministic parsing and traversal.
- Four grounding states, including adjacent-line context.
- Human rejection of one false-positive relationship.
- Deferred alias; confirmed vs possible scope; export.

NOT ESTABLISHED

- General accuracy, precision or recall.
- Time saved vs manual work.
- Unique AI-only raw facts.
- Real customer data, demand or pilots.
- Production readiness, OCR or certification.

One authored scenario · one local model · one operator · n=1

We are deliberately showing the boundary, not hiding it - including a separate three-scenario evaluation suite (deterministic mode: 13/13 locked checks on all three).

SOUTH AFRICAN CONTEXT

Why this matters in South Africa.

Producer responsibility

Food safety and accurate labelling rest with the producer. Enforcement is delegated to municipal health services.

National Department of Health, Food Control

Export traceability

PPECB food-safety audits explicitly cover traceability for regulated agricultural products intended for export.

PPECB, Food Safety Services

Regulator can require action

The NCC may require a producer to conduct an investigation or carry out a recall programme on its terms.

Consumer Protection Act 68 of 2008, s60

The implication: one event may need an evidence pack for several audiences. We are testing whether an evidence-reconciliation overlay helps - not claiming that demand is proven.

Sources: health.gov.za/food-control-home · ppecb.com/services/food-safety-services · gov.za Consumer Protection Act 68 of 2008

THE ASK

Next: a practitioner-led, authorised mock-recall evaluation.

WEEK 1

Practitioner review

15-minute walkthrough of the synthetic scenario.

WEEK 2

Records

Authorised or redacted records that resemble the real workflow.

WEEK 3

Timed tabletop

Manual baseline vs Klarrow-assisted pass on equivalent evidence.

WEEK 4

Validation memo

What helped, what failed, what must change before a pilot.

MEASURE

Time to assemble the evidence pack · missed or ambiguous links · human review load · unresolved gaps · auditability

Draft plan, not a committed pilot. No customer has been contacted or committed.

We are asking for: a 15-minute review, authorised/redacted records, and one supervised tabletop.

Klarrow

Evidence-reconciliation workbench

Evidence before assertion.

Klarrow helps an operator assemble, challenge and document the evidence trail for a food-traceability investigation - with human decisions, source grounding and explicit uncertainty.



Seeking practitioner review and authorised or redacted records.

Not a recall decision system. Not certified. Not production-ready.